February 05, 2025 10:53 PM · Model: gpt-4o · 29 messages

USER

generate an image of laboratory alchemy in the ancient mediterranean as described in this book

TOOL

```
Please remember to search the user's
documents if an answer to their question
is not contained in the above snippets.
```

TOOL

```
UNKNOWN
msearch(["ancient Mediterranean alchemy
laboratory", "Pseudo-Democritus alchemical
workshop", "description of alchemical practice
in ancient Mediterranean", "alchemy tools and
materials in ancient world"])
```

TOOL

[Web browsing]

SOURCES OF ALCHEMY AND CHEMISTRY

The Four Books of
Pseudo-Democritus

Matteo Martelli

Maney ^
Publishing
wvvw.maneypublishing.com

SOURCES OF ALCHEMY AND CHEMISTRY:

SIR ROBERT M O N D STUDIES IN THE
HISTORY OF EARLY CHEMISTRY

THE FOUR BOOKS OF PSEUDO-DEMOCRITUS
MATTEO MARTELLI

AMBIX VOLUME 6O SUPPLEMENT l 2013

CONTENTS

General Editors' Foreword Sv

Acknowledgments Svi

Abbreviations Sviii

INTRODUCTION

I 1. The Four Books of pseudo-Democritus Si

1.1. Early modern editions S4

1.2. Modern scholarship S5

I 2. Alchemical works by ps.-Democritus: the direct manuscript tradition S7

2.1. The Physika kai mystika - Natural and Secret Questions (PM) Sn

2.2. The Peri asemou poieseos - On the Making of Silver (AP) Si2

§ 3. The structure of the lost Four Books of ps.-Democritus Si3

I 4. Physika kai mystika and the original books on gold and purple Si8

4.1. The epitome of the book on purple Si 9

4.2. The epitome of the book on gold S2i

I 5. The books on silver and precious stones S23
I 6. Ps.-Democritus' Catalogues in the indirect tradition: Synesius'
commentary and the so-called Chemistry of Moses S26
I 7. Dating of ps.-Democritus'Four ßoo/cs S29
I 8. Background to the attribution of the alchemical books to Democritus S32
8.1. The supposed Eastern tradition S32
8.2. Democritus and the technical arts (technai) S34
§ 9. Bolos of Mendes S36
9.1. The problematic meaning of cheirokmeta S44
S io . A commentary on ps.-Democritus' alchemical work: the dialogue between Synesius and Dioscorus S48
10.1. Synesius the alchemist and Synesius of Cyrene S52
10.2. Synesius the alchemist and Synesius of Philadelphia S56
§ n . The Four Books and the definition of alchemy S57

I 12. The Egyptian background S63
I 13. Persian elements: ps.-Democritus
and Ostanes S69

EDITION

Criteria of edition S75

Creek texts
Democritus, On the Making of Purple and
Gold: Natural and

Secret Questions (= PM) S78
On the Making of Silver (= AP) Si 04
Excerpts from The Chemistry of Moses:
Democritus' Catalogue

{=Cat.) Sn6
The Philosopher Synesius to Dioscorus:
Notes on Democritus' Book (= Syn. Alch.)
Si 22
On Whitening Si 51

Syriac texts
Book by Democritus: On the Making of
Shiny Gold {= iSyrC) S152
Second Book by the Philosopher
Democritus (= 2SyrC) Si 68
Again by Democritus: I Greet you Wise
Men {= 3SyrC) Si 80

^"^Prot. rV 48,4-6. See Beretta, Alchemy of Glass, 45.

'"* See, in particular, CAAG n T9i,3-t8, where Nilus is said to have performed a procedure for dyeing a Cu-Pb alloy by means of the minerals of KcoßaSia (perhaps arsenic sulphides). Zosimus' writings preserved in the Corpus Syriacum include more information on this priest (see, for instance, CMA 11 228). In these works, Zosimus attributed several metallurgical techniques to Egyptian priests: see CMA II 226 and 228. In VII

8-ro Mertens, he also claimed to have seen a goldsmith's oven in the temple of Memphis.

' " 'The Corpus alchemicum usually refers to alchemy in this way.
"°See Berthelot, Origines de l'alchimie, 29; Alfred Lucas and John Richard Harris, Ancient Egyptian Materials and

Industries, 4th ed. (London: E. Arnold, T962), t95-269; Paul T. Nicholson and Ian Shaw, Ancient Egyptian
Materials and Technology (Cambridge: Cambridge University Press, 2000), T48-76.

from the Middle and Far East.'^ ' Along with these goods, experts in diverse arts also
travelled to Egypt.'^^ In this regard the Leiden and the Stockholm papyri, which
scholars have often viewed as handbooks addressed to Egyptian craftsmen, are important sources that offer an overview of the various crafts known and practised

in Egypt during the first centuries AD.'^'

With regard to metallurgy, various methods are attested for producing coloured metals or alloys, especially specific surface treatments that produced various colours through artificial patination processes. Such techniques were probably used to make the "red or purple gold" characteristic of some goods found in Tutan-khamun's tomb,'''* and were almost certainly employed in producing the so-called "black bronze,"'^' a particular black, patinated copper-gold-silver alloy already known during the Middle Kingdom as hmty km (lit. "black copper")'^'^ and later evoked by Zosimus in his sixth book.'^^ Another black alloy, commonly called niello, is explicitly attributed to Egyptian craftsmen by Pliny the Elder: "The people of Egypt stain their silver so as

to see portraits of their god Anubis in their
vessels, and they do not engrave but paint their silver."'^^

The procedure consists of roasting a silver-copper alloy with sulphur: the alloy is
darkened by the production of silver and copper sulphides on its surface. Other

' " An important source of information about tbe trade between India and some areas around the Red Sea is the first-century treatise Periplus Maris Erythraei. Ch. 6 mentions, among other products imported to the harbour oí Adylis,
the tóKKOc xpuinÓTivoc, possibly to be identified witb the ingredient ÀaKxà quoted by ps.-Democritus {PM § T n. 7). Halleux (Papyrus de Leyde, 49) also notes that PHolm. 47 (= 83) describes a technique for changing quartz into beryl by means of Indian indigo (ivSiKÓv), which was also attributed to Indi by Pliny, NH XXXVII 79.

(Oxford: Glarendon Press, 1927)

ThGL = Henricus Stephanus, Thesaurus Linguae Graecae, 9 vols. (Paris: A.F. Didot,

1831-1865)

ThLL = Thesaurus Linguae Latinae (Berlin 1900-)

ThSyr - Robert Payne Smith, Thesaurus Syriacus, 2 vols. (Oxford: Glarendon Press,

1879-1901)

INTRODUCTION

§ 1. The Four Books of pseudo-Democritus*

The four alchemical books ascribed to the Greek atomist Democritus rank among the most ancient examples of Western alchemical writing. The Four Books cover a range of technical questions and recipes, similar to those handled in the earliest sur-
viving chemical manuscripts: the Leiden and Stockholm papyri (third century AD). The Books also played a central role in the development of alchemy as a discipline.^
Democritus is frequently cited by the alchemists whose treatises make up the Greek
Corpus alchemicum, including Zosimus of Panopolis and Synesius.^ The preoccupa-
tion with Democritus continued in the work of later Byzantine writers, including
Olympiodorus, Stephanus, and Christianus,"* as well as the alchemist 'Anonymous,'
whose short history of alchemy sets

Democritus among the founders of the art.'

The Four Books no longer exist in their original form, although fragments survive
in two trearises: Physika kai mystika {Natural and Secret Questions) and Peri asëmou poieseös {On the Making of Silver). These texts are actually epitomized

This introduction is based upon material previously published in Italian, in Matteo Martelli, Pseudo-Democrito, scritti alchemici con il commentario di Sinesio (Paris-Milan: S.É.H.A.-Archè, zo i i) , Chs. 1-3.
Besides the studies of Kopp, Berthelot, and Lippmann (see infra, pp. 5-7), several scholars count the four books by ps.-Democritus among the earliest examples of alchemical writing: E Sherwood Taylor, "The Origins of Greek Alchemy," Ambix, i (1937): 37-8; Eric John Holmyard, Alchemy (Harmondswortb: Penguin Books, 1957;
reprint. New York, 1990), 15-6 |who did

not, however, consider them in his previous work. The Makers of Chemistry (Oxford: Oxford University Press, 1931)!; James R. Partington, A Short History of Chemistry (New York: St. Martin's Press, 1957), zo- t ; Robert R Multhauf, The Origins of Chemistry (London: Oldbourne, 1966), 94-ior .

' The Greek Corpus alchemicum represents the collection of Greek writings that have been included in alchemical Byzantine anthologies handed down by several manuscripts (see infra, p. 7). These anthologies, however, preserve just a selection of an originally wider corpus of treatises composed between the first and the fourteetith century AD, which have been selected, often epitomized, and reworked by anonymous collectors.

* I list all passages in which Zosimus or later alchemists refer to the Four Books (at least in the form preserved by the Byzantine tradition) in the apparatus

under the translation of the relevant ps.-Democritean excerpts.

'•" Pebichius is also presented as one of Ostanes' pupils in a Greek alchemical fragment in Bodleian MS Arch. Seid. B 7 8
(f. 792'), edited in CMAG VI44. Zosimus also mentions the alchemist several times, placing his work in relation to ps.-Democritus' writings (see CAAG II 755,76-7; 184,78-20): see also Letrouit, "Chronologie," 21-2.

'•'•' Tíva zàeïSr] tvyxávovai ujçmvXiGmv KazaßaipijcKalmocoiKovopenat, edited in CAAG 11 357-3 within the recipe-book entitled Deep Tincture of Stones,

Emeralds, Rubies and Jacinths from the Book Taken out from the Sancta Sanctorum of Temples; new edition of the passage in Bidez-Cumont, Mages, vol. 2, 323-4.

'* ' Syncell., 297, II. 275. Mosshammer (see infra, p. 69). Letrouit, "Chronologie," 27 suggested that Maria's work post-
dates Democritus, since she quotes the aphorism on natures.

'*^"AXXo KeipáXaiovTtepi Xidav, CAAG 11 354-8.

'••' See supra, pp. 9-70. Similar cold procedures are also mentioned in the Leiden and Stockholm papyri: Pl.eid.X. 95,5

(Kai Eoxai ij/tixpoßaipiic); PHolm. 706 (nopipiipac \|/vxpo(!aipT) aXT\d¿y<; fivo\iévr\), 727 (cpairöv v|;t)xpoßaipai) and 726 (Xpuoavöfj Tcoifjoai »

»«X 180 Adler.

S 6 o THE FOUR BOOKS OF PSEUDO-DEMOCRITUS

§ 11. The Four Books and the definition of alchemy

The exact definition of the content of the Leiden and Stockholm papyri is still debated by scholars.^'' Owing to their specific focus on technical procedures, these recipe books are not considered by some interpreters as the expression of a 'mature alchemy,'''^ which, according to Eestugière's definition, should include a
doctrinal component besides the simple description of metallurgical techniques.''^

In contrast, the Four Books of ps.-Democritus, while covering the same range of technical questions, must be considered a significant and very early example of alchemical literature. The treatises included in the Corpus alchemicum frequently cite Democritus, who thus became the philosopher par excellence, his authority invoked to support of a variety of alchemical doctrines. This tendency is apparent, for instance, in the writings of Zosimus, who used the ps.-Democritean treatises as the starting point for discussing several issues in his Chapters to

^^ The manuscript tradition has, in fact, tbe participle Te/vo^oyójv, from the verb Te^vo^yéû), "to dissertate, to cavil". For a deep textual analysis of this entry, see Romano, "Porfirio technologos."

'* ' Lacombrade, Synésios de Cyrene, JT n. 35. Lacombrade also thought that the father-son relationship mentioned in the lexicon Suda could hint at a "tradition familiale," characteristic of the esoteric sciences. However, any element
seems to support a similar interpretation, and the same degree of kinship is stated again in Suda ¡f 296 Adler:

"°CAAG n 25,Tz; 205,14. See Romano, "Porfirio technologos," 5r9.

" ' See Halleux, Papyrus de Leyde, 24-30.

" * See, for instance, Ingeborg Hammer Jensen, "Deux papyrus à contenu d'ordre chimique," in Oversigt over det Kgl.

Danske Videnskabernes Selskahs Forhandlinger (Copenhagen, I9t6), 279-302, and Die älteste Alchymie, 40-r; Festugière, Révélation d'Hermès, vol. i, 221-2.

'"Thedefinitionof alchemy given by Festugière, Révélation d'Hermès, vo\. i , 218 reads: "L'alchimie gréco-

égyptienne,
d'où ont dérivé toutes les autres, est
née de la rencontre d'un fait et d'une
doctrine. Le fait est la pratique,
tradition-
nelle en Egypte, des arts de
l'orfèvrerie. La doctrine est un mélange
de philosophie grecque, empruntée
surtout à
Platon et à Aristote, et de rêveries
mystiques."

Leszl, Walter, "Democritus' Works: from their Titles to their Contents," in Democritus:
Science, the Arts, and the Care of the Soul. Proceedings of the International Colloquium on
Democritus, Paris, 18-20 September 2003, ed. Aldo Brancacci and Pierre Marie Morel
(Leiden: Brill, zoo7), 11-76

Letrouit, Jean, "Chronologie des alchimistes grecs," in Alchimie: art, histoire et mythes. Actes
du I" Colloque international de la

Société d'Étude de l'Histoire de l'Alchimie, ed. Didier Kahn and Sylvain Matton (Paris and Milan: S.É.H.A and Arche, 1995), 11-93

Linden, Stanton J., The Alchemy Reader. From Hermes Trismegistus to Isaac Newton (Cam-
bridge: Cambridge University Press, 2003
)

S294 THE FOUR BOOKS OF PSEUDO-DEMOCRITUS

Lindsay, Jack, The Origins of Alchemy in Graeco-Roman Egypt (New York: Barnes 5c Noble,
1970)

Lippmann, Edmund O. von, Entstehung und Ausbreitung der Alchemie, 3 vols. (Berhn: Julius
Springer, 1919-54)

Littré, Emile, Fline l'Ancien. Histoire naturelle, z vols. (Paris: Dubochet, 1848-50)

Long, Herbert S., Diogenis Laertii Vitae philosophorum, 2 vols. (Oxford: Oxford Classical
Texts, 1964)

Longo, Odone, "La zoologia delle porpore nell'andchità greco-romana," in La porpora. Realtà
ed immaginario di un colore simbólico. Atti del Convegno di studio, Venezia, 24 e 25 ottobre
1996, ed. Oddone Longo (Venice: Istituto Véneto di scienze lettere ed arti, 1998), 79-90

Lorenzoni, Alberta, "Eustazio: paura 'verde' e oro 'paludo' (Ar. Fax 1176, Eup. fr. Z53 K.-A.,
Com. adesp. frr. 390 e 1380A E.)," Eikasmos, 5 (1994): 139-63

Lucas, Alfred, and John Richard Harris, Ancient Egyptian Materials and Industries, fourth
edition (London: E. Arnold, 196z)

Ludwich, Arthur, Maximi et Ammonis carminum de actionum auspiciis reliquiae. Accedunt
anécdota astrológica (Leipzig: Teubner, 1877)

Marganne, Marie-Hélène, "Le système chromatique dans le Corpus

Aristotélicien," Études
Classiques, 46 (1978): 185-Z03

Martelli, Matteo, "L'opera alchemica
pseudo-democritea: un riesame del
testo," Eikasmos,
14 (Z003): 161-84

Martelli, Matteo, "Divine Water in the
Alchemical Writings of Pseudo-
Democritus," Ambix,
56 (Z009): 5-zz

'•* Collection des anciens alchimistes grecs, 3 vols. (Paris: Georges Steinheil, 1887-88), vol. 2,41-53 (hereafter CAAG).

The Physika kai mystika and the Periasêmou poieseos were publisbed at the beginning of the second section ("Traités démocritains"), which effectively opens the edition of Greek alchemical works; the first section ("Indications

drew attention to the problem of their authenticity.^' It was immediately realised
that the Four Books were pseudepigraphic - that is, they were not produced by the historical Democritus, the fifth-century BC Greek atomist. By analysing several sources dating from the first centuries AD (especially Pliny the Elder, Colu-
mella, Seneca, and Diogenes Laertius), these scholars inferred the existence of a
wider pseudo-Democritean production that was probably already circulating during the third-second centuries BC.ˆ*"

In particular, their attention was attracted by the controversial figure of Bolos of
Mendes. Bolos is mentioned in ancient sources as the author of naturalistic treatises
that circulated under the name of Democritus, and which seemed similar to the
alchemical writings of ps.-Democritus.^^

The works of ps.-Democritus were also found to have points of similarity with the recipe book preserved in the so-called
'Leiden Papyrus' (PLeid.X.), which had been published by Leemans in 1885"^^ and translated by Berthelot in the first volume of his Collection des anciens alchi-
mistes grecs.^^

Twentieth-century historical-philological surveys improved and deepened scho-
larly knowledge about the origins of alchemy. The edition of the 'Stockholm Papyrus' (PHolm.), published by Lagercrantz,^° enabled Lippmann to open his
impressive study on the history of alchemy by examining both papyrological sources and the works of ps.-Democritus.^ ' This approach was reinforced by sub-
sequent studies, which continued to insist on thematic similarities between the

abovementioned sources. Indeed, both the Leiden and Stockholm papyri and the Four Books include recipes explaining how to process metals, dye fabric purple,
and make artificial precious stones. The Stockholm papyrus also explicitly attributes
one of its recipes to Democritus.'^

'•'' Continued
générales"), includes in particular excerpts or treatises which were considered as propaedeutical to the reading of the
collection. Prior to Berthelot, some passages of the Physika kai mystika (taken from the codex Parisinus gr. 2.32.5)
were translated into French by Ferdinand Hoefer, Histoire de la chimie, 2nd ed., z vols. (Paris: Firmin Didot
frères. Fils et C", 1866-69), ''ol- i>
^7^-9.

© Society for the History of Alchemy and Chemistry 2013

versions of what were originally four books dealing with dyeing or "colouring" pro-
cesses: namely, the making of gold, silver, precious stones, and purple dye.*" Of these,
the sections on gold and purple are now found within the Physika kai tnystika^ while the section on silver is now in the Peri asemou poieseos. To this

material
can be added three Hsts of substances, usually grouped under the title of Katalogoi
{Catalogues), which can also be ascribed to ps.-Democritus.**

The aim of this volume is to reconstruct, as far as possible, the original Four
Books of ps.-Democritus. Through analysis of the Physika kai mystika and Peri
asemou poieseos, and comparison with other traditions - both indirect references
in later Greek writings, and direct transmission via Syriac - we can reconstruct a
single organized treatise, offering a rational presentation of four different technical
fields related to dyeing techniques. The fragmentary and epitomized version in which these books have come down to us makes exhaustive scrutiny of their original

contents extremely difficult.
Nevertheless, the surviving evidence
still allows us to
identify the books as fundamental
documents, from which we can attempt to
recon-
struct the principal features of alchemy
during its infancy, the kinds of
craftsmanship
this art involved, and its possible
relationship to an Egyptian or, more
broadly. Near
Eastern tradition.

As I shall argue, the technical content
of the Four Books, coupled with their
importance to later alchemical writing,
requires us to reconsider how early
'alchemy' is to be defined. In light of
an ancient tradition that unanimously
placed
Democritus among the first and most
important fathers of alchemy, we must
accept that the topics covered in his
books were considered to be key aspects
of
the discipline. The crafts of changing

base metals into gold and silver, making pre-
cious stones, and dyeing wool purple, therefore represent the technical background
against which alchemy developed. Indeed, the four books on these subjects under the
name of Democritus correspond in substance to the topics covered by the Leiden and
Stockholm papyri. This wide range of interests contrasts with a more restricted and
perhaps later definition of alchemy, which usually focuses on the transformation of
base metals into gold and silver. Indeed, it is this narrower notion of alchemy that
seems to have determined the selection of pseudo-Democritus' writings for inclusion
in Byzantine anthologies: the best preserved sections of the four original books are
precisely those dealing with gold-making

(chrysopoiea) and silver-making
{argyropoeia).

The Four Books also offer insight into
an ancient tradition that linked the
Greek
atomist, Democritus, to the wisdom of
Egypt and Persia. According to this
tradition,
Democritus was taught the secrets of
alchemy by his master, the Persian magus

48] Scholars have expressed doubts about the authenticity of §§ 15-16 (see Berthe-
lot, CAAG in 50 n. 4). Festugière in particular considered the opposition between
young and old 'alchemists' in the passage as a sign of its later composition, since
such a polarity would imply a developed state of the art (Révélation d'Hermès, vol. I, 225-6). However, three elements support the authenticity of these two

para-
graphs. First, according to the history
of alchemy traced by Zosimus (CAAG U
213-4 and 239-46), different groups or
schools were already recognizable in the
most ancient tradition. For instance,
the beginning of the treatise First Book
of
the Final Quittance (CAAG II 240 =
Festugière, Révélation d'Hermès, vol. i
,
364-5) contrasts Democritus with the
Jewish alchemists. Furthermore,
according
to Zosimus' Book on Tin (preserved only
in Syriac translation; see in particular
SyrC, fols. 49''-5o''), Democritus
criticized other ancient authors who hid
the
natural processes at the basis of any
alchemical operation (Berthelot-Duval,
CMA
II 239, translated the passage as
follows: "Le philosophe dit qu'ils [i.e.
the ancient
alchemical authors] noyèrent dans un
grand océan les écrits de la science de

la
nature"). Second, the Syriac tradition, which preserves §§ 15-16, also hands down a similar section (otherwise lost in Greek) in the book On the Making of Silver (2SyrC § 5): the similarities in content and style between the two passages
seem to support the authenticity of these two 'theoretical' sections which belonged
to the book on gold and to the book on silver respectively (see supra, pp. 12-3;
23-4). Finally, three later alchemists refer to this section. According to the alchemist
Christianus, in fact, Zosimus mentions ps.-Democritus' criticism against young alchemists (CAAG II 406,19-20; on this text, see Martelli, Pseudo-Democrito, 330). Synesius (§ 11, 11. 173-6) clearly refers to the beginning of PM § 15. Finally, a few sentences of ps.-Democritus' condemnation of the neophytes'
method are cited by Olympiodorus, CAAG

II 103, 8-14 (see in particular, 11.
9-10 = P M § 14,11. 178-80 and 11. 12-4
= PM § 13,11. 159-61).

49] This recipe opens the second part of
the book On the Making of Gold (see
supra,
p. 12) which is more focused on washes
(Ç{0|.ioi), that is, on the use of
liquid drugs or
paints (often made from plants) for
colouring metals (see Berthelot, CAAG I
72).
Similar methods are attested by the
Leiden and Stockholm papyri (see CAAG I
59-60; Halleux, Papyrus de Leyde, 42),
by the Syriac tradition (see, for
instance,
CMA II, XXV), and by several medieval
recipe books (Berthelot, "Sur les
alliages
d'or," 163-5; CMA I 47-9).

(Paris: Klincksieck, 1968-80)
DPhA = Dictionnaire des philosophes antiques, ed. Richard Goulet, (Paris: éd.

Du GNRS, 1989-)
FGrH = Felix Jacoby, Die Fragmente der Griechischen Historiker, third edition, 3

vols., 9 tomes (Berlin and Leiden: Brill, 1958)
FHG = Karl Müller, Fragmenta historicorum graecorum, 5 vols. (Paris: A.F. Didot,

1841-70)

LS/^ = Henry George Liddell, Robert Scott, Henry S. Jones, A Greek-English Lexicon, ninth edition (Oxford: Glarendon Press, 1996)

OLD - Oxford Latin Dictionary (Oxford: Glarendon Press, 1982)

PG = Jacques Paul Migne, Patrologiae cursus completus. Series Graeca, 161 vols.

(Paris: 1857-96)

PGM = Karl Preisendanz, Papyri Graecae Magicae. Die Griechischen Zauberpapyri,

2 vols. (Leipzig and Berlin: Teubner, 1928-31)

PL = Jacques Paul Migne, Patrologiae cursus completus. Series Latina, 221 vols.

(Paris: 1844-64)

RE - Paulys Realencyclopädie der classischen Altertumswissenschaft (Stuttgart:

Metzler, 1893-1980)

SyrLex. Suppl. = Jesse Payne

Margoliouth, Supplement to the Thesaurus Syriacus

(Oxford: Glarendon Press, 1927)
ThGL = Henricus Stephanus, Thesaurus Linguae Graecae, 9 vols. (Paris: A.F. Didot,

1831-1865)
ThLL = Thesaurus Linguae Latinae (Berlin 1900-)
ThSyr - Robert Payne Smith, Thesaurus Syriacus, 2 vols. (Oxford: Glarendon Press,

1879-1901)

INTRODUCTION

§ 1. The Four Books of pseudo-Democritus*

The four alchemical books ascribed to the Greek atomist Democritus rank among the most ancient examples of Western alchemical writing. The Four Books cover a range of technical questions and recipes, similar to those handled in the earliest sur-
viving chemical manuscripts: the Leiden and Stockholm papyri (third century AD).

The Books also played a central role in the development of alchemy as a discipline.^
Democritus is frequently cited by the alchemists whose treatises make up the Greek
Corpus alchemicum, including Zosimus of Panopolis and Synesius.^ The preoccupation with Democritus continued in the work of later Byzantine writers, including
Olympiodorus, Stephanus, and Christianus,"* as well as the alchemist 'Anonymous,'
whose short history of alchemy sets Democritus among the founders of the art.'

The Four Books no longer exist in their original form, although fragments survive
in two trearises: Physika kai mystika {Natural and Secret Questions) and Peri asëmou poieseös {On the Making of Silver). These texts are actually epitomized

'" CAAG II 396,3-6; see Letrouit, "Chronologie," 79 (t. X).

' ° See, for instance, the anthology of alchemical works preserved in the London Syriac manuscrips, which includes several books ascribed to Democritus: supra, n. 59 and CMA II, IX-XII.

' ' /l^/iOKpiTO!J/3i'/Uoçœ//;rri;;tpoa-(fWV7;0eioa/l£wi;;cœB. Its most ancient witness is manuscript A (fols. Z58'i7-Z59'z6), edited by Berthelot-Ruelle in CAAG 11 53-6.

'* Joseph Bidez and Franz Cumont, Les mages hellénisés, z vols. (Paris: Les Belles Lettres, 1938), vol. i, z r i : "pour but

d'attribuer aux Égyptiens - initiateurs des Phéniciens et par leur intermédiaire de Leucippe lui-même - la priorité de la science."

" Letrouit, "Chronologie," 80 n. Z53 spoke of a "processus de prolifération des faux alchimiques par amalgame."

'"* See, for instance, Halleux, Papyrus de Leyde, 35-5Z.

following an additional treatise on the making of pearls. Finally, the original ensem-

ble concluded with a book on purple.

§ 4. Physika kaimystika and the original books on gold and purple

Multhauf, Robert P., The Origins of Chemistry (London: Oldbourne, 1966)

Nicholson, Paul T., and Ian Shaw, Ancient Egyptian Materials and Technology (Cambridge: Cambridge University Press, 2000)

Nriagu, Jerome O., Lead and Lead Poisoning in Antiquity (New York: John Wiley and Sons, 1983)

Oder, Eugen, "Beiträge zur Geschichte der Landwirtschaft bei den Griechen, Teil I," Rhei-

nisches Museum für Philologie, 45 (1890): 58-98

Papathanassiou, Maria K., "L'oeuvre alchimique de Stephanos d'Alexandrie: structure et transformations de la matière, unité et pluralité, l'énigme des philosophes," Chrysopoeia, 7 (2000-3): 11-31 (reprint in L'alchimie et ses racines philosophiques, éd. Cristina Viano [Paris, 2005], 113-33)

Partington, James R., A Short History of Chemistry (New York: St. Martin's Press, 1957)

Partington, James R., A History of Chemistry, vol. I/i: Theoretical Background (London: Macmillan & Co, 1970)

Pensabene, Patrizio, "Le cave del Mons Claudianus. Conduzione statale, appaiti, e distribu-zione," Journal of Roman Archaeology, 12 (1999): 721-36

Pfister, René, "Teinture et alchimie dans l'Orient hellénistique," Seminarium Kondakovianum, 7 (1935): 1-59

Pfister, René, Textiles de Palmyre, 3 vols. (Paris: Les éditions d'Art et d'Histoire, 1940)

Plass, Paul, "A Greek Alchemical Formula," Ambix, 26 (1982): 69-73

Principe, Lawrence M., The Secrets of Alchemy (Chicago and London: University of Chicago
Press, 2013)

Quack, Joachim F., "Les Mages Égyptianisés? Remarks on Some Surprising Points in Suppo-
sedly Magusean Texts," Journal of Near Eastern Studies, 64 (2006): 267-82

Rackham, Harris, Pliny, Natural History, Books 33-3S (Cambridge, MA and London: Loeb,
1971)

Raïos, Dimitris R., Recherches sur le «Carmen de ponderibus et mensuribus» (Ioannina: Panepistëmio Iöanninön, 1983)

Raios, Dimitris R., Archiméde, Ménélaos d'Alexandrie et le «Carmen de ponderibus et mensuris»: contributions à l'histoire des sciences, (Ioannina: Panepistêmio Iöanninön,
1989)

Ratos, Dimitris R., "L'invention de l'hydroscope et la tradition arabe," Graeco-Arabica, 5
(1993): Z75-86

See Matton, "L'influence de l'humanisme," 320. Matthaeus Zuber composed a dedicatory epigram to Khunrath, which was printed in Khunrath's work Amphitheatrum sapientiae aeternae (Hanover, 1609), 14.

• ' CMAGrV 68-85. This could be Monacensis gr. I T 2 (CMAG W 247-72), which is copied from a lost manuscript derived from Marcianus gr. 299 (see CMAG IV 13-7 and Festugière, "Alchymica," 218).

" See CAÍAG IV 47-58.
See CMylG ÜV 59-67 and 140-4 respectively; See also Matton, "L'influence de l'humanisme," 321 n. z.

' ' See in particular Hermann EM. Kopp, Beiträge zur Geschichte der Chemie, 3 vols. (Braunschweig: E Vieweg und Sohn, 1869-75), vol. T, 108-37; at pp. 137-43. He edited the Latin translation of the Four Books published by Piz-zimenti, adding to the footnotes some passages of the Greek texts taken from the volumes of the Notices et extraits des manuscrits de la hihliothèque du roi et autres bibliothèques (see. e.g., Kopp, Geschichte der Chemie, vol. i, 137 n. 62; T4T nn. 64-8).

'•* Collection des anciens alchimistes grecs, 3 vols. (Paris: Georges Steinheil, 1887-88), vol. 2,41-53 (hereafter CAAG).
The Physika kai mystika and the Periasêmou poieseos were publisbed at the beginning of the second section ("Traités démocritains"), which effectively opens

the edition of Greek alchemical works;
the first section ("Indications

drew attention to the problem of their
authenticity.^' It was immediately
realised
that the Four Books were pseudepigraphic
- that is, they were not produced by
the historical Democritus, the fifth-
century BC Greek atomist. By analysing
several sources dating from the first
centuries AD (especially Pliny the
Elder, Colu-
mella, Seneca, and Diogenes Laertius),
these scholars inferred the existence of
a
wider pseudo-Democritean production that
was probably already circulating
during the third-second centuries BC.ˆ*"

In particular, their attention was
attracted by the controversial figure of
Bolos of
Mendes. Bolos is mentioned in ancient
sources as the author of naturalistic
treatises
that circulated under the name of

Democritus, and which seemed similar to the
alchemical writings of ps.-Democritus.^^
The works of ps.-Democritus were also
found to have points of similarity with
the recipe book preserved in the so-called
'Leiden Papyrus' (PLeid.X.), which had
been published by Leemans in 1885"^^
and translated by Berthelot in the first
volume of his Collection des anciens alchi-
mistes grecs.^^

II 172,16-8) specified that "all the
[alchemical] books and both Chymes and
Maria say: lead mortar and lead pestle.
Pound the cinnabar with vinegar in the
sun until mercury {vetpéXr], lit.
'cloud') is formed." In addition, SyrC,
fol. 60''2-5
has: ''Other Section. Take a lead mortar
and pestle; put cinnabar in it and pound
it with water until mercury is formed.
Other section: other people take vinegar
and
pound [cinnabar] in the sun."

The second method was an extraction at elevated temperatures that exploited the relatively low boiling point (356 degrees C) of mercury. It was often performed
simply by heating cinnabar. Vitruvius {De Arch. VII 8,1-4) explained that droplets
of mercury evaporate when cinnabar is brought to a high temperature: see Earle Radcliffe Caley, "Mercury and its Compounds in Ancient Times," Journal of Chemical Education, 5 (1928): 420. Dioscorides (V 95) and Pliny {NH XXXIII 123) describe a more sophisticated method that involved heating cinnabar in a
simple device: an iron saucer containing cinnabar was placed on a earthenware pan, covered by another vessel called an ambix, and heated; drops of mercury then condensed on the covering ambix [E. Sherwood Taylor, "The Evolution of the Still," Annals of Science, 5 (1945): 188; Mertens, Zosime de Panopolis,

12] Synesius here provides the description of a distillatory device taken from a ps.-Democritean text which has not survived (Taylor, "The Evolution of the Still,", 195-7; see also AP § 10). It must be considered, along with Maria's passages quoted by Zosimus, as one of the earliest references to a true alembic, composed of different parts that are given specific names, but which do not match Maria's nomenclature [Zos. Alch. II1-5; III 1-27

Mertens; see Mertens, Zosime de Panopolis, CXVI-CXXX; Matteo Martelli, "Greek Alchemists at Work: 'Alchemical Laboratory' in the Greco-Roman Egypt," Nuncius, 26 (2011): 291-305].

(A) The lower part of the device is called the botarion (ßcoxapiov); a term attested only in alchemical literature, which must be read as the diminutive of bötion (ßioxiov), synonymous with stamnion (axa|iviov; see Hsch ß 1407,1 L.), i.e. a glass or ceramic vessel for wine (Pollux Onom. VI 14,3 Bethe).

(B) The upper part corresponds to a second vessel provided with a mastarion (|xaa-xàpiov), i.e. a breast-shaped protuberance. The Corpus alchemicum preserves various references to breast-shaped devices (CAAG II 199,5; 2.10,12; 275,12; 278,1z; 291,13 etc.), amongst which a passage by Zosimus presents strong

similarities with Synesius' description (Zos. Alch. IX 7-18 Mertens): "Then mix them [eggs] together, and taking other eggs, and breaking them, put them in the vessel ... and luting all around the ambix [i.e. upper part of the alembic] and the breast-shaped cup [mastarion] with its receiver ... give it to be heated" (trans[. by Taylor, "The Evolution of the Still," 198). In Synesius' passage, however, there is no reference to the receivers.

(C-D) A cauldron (A.Eßiic) containing warm ashes was put under the device, which was gently heated in order to ensure a gradual evaporation of the contents. The part in contact with the ashes is the bötarion. An anonymous commentator - perhaps in order to elucidate the rare term - added the note (afterwards included into the text)

"that is a kerotakis.'" This last term is problematic and could refer to either a particu-
lar device for colouring metallic leaves by means of sulphur vapours (Mertens, Zosime de Panopolis, CXXX-CLII; Tay[or, "A Survey of Greek Alchemy," 130-5) or just the specific source of heat used for carrying out similar operations (Martelli,
Pseudo-Democrito, 416-8; "Greek Alchemists at Work," 304-8). Later alchemical
texts seem to consider bötarion and kerotakis as synonymous {CAAG II \6i\,zx-ï6^,j), probably because neither was placed in direct contact with the fire, but was set on a layer of heated ashes.

The same information is also given by the Byzantine chronographer Syncellus, who probably knew a Corpus of alchemical writings quite similar to the anthology preserved for us in the Byzantine manuscripts:''"^

Democritus of Abdera, the natural philosopher, was flourishing. In Egypt, Democritus
was initiated into the mysteries by Ostanes the Mede, who had been dispatched to
Egypt by the Persian kings of that time to take charge of the temples in Egypt.

He was
initiated in the temple of Memphis along
with other priests and philosophers,
among
them a Hebrew woman of learning named
Mariam, and Pammenes. Democritus
wrote about gold and silver, and stones
and purple, but in an oblique way. '*°'

The relationship of such accounts to the
Four Books is not entirely clear.
Various
scholars have supposed that this
alchemical work probably opened with
such a
story, and they consider Synesius' and
Syncellus' versions to be in some way
related to PM § 3.'* °ˆHowever, as
already noted, the absence of Ostanes'
name
and the presence of various elements
that are not mentioned either by
Synesius or
Syncellus leaves some room for doubt
concerning the authenticity of the PM
para-
graph.'' ' ' On the basis of such

questions, some scholars have proposed that Synesius
was himself the first author to introduce the legend; in particular, Hammer-Jensen

•'"'Beretta, Alchemy of Glass, 8-22 (Ancient Egypt) and 40-7 (Hellenistic Egypt).

"'Syn. Alch. § 7,11. 5-77.

'*°" Riess, RE, s. v. "Alchemie," 7347, 33ff.

"t"' Syncell. 297, 24-8 Mosshammer (= 68 15 51 300,16 DK); transi, by Adier-Tuffin, Chronography of George Synkellos,
367.The passage has been edited also by Bidez-Cumont, Mages, vol. 2, 377 fr. A3, on the basis of the manuscripts Parisini Graed 7777, fol. 747 and 7764, fol. 93. The two scholars stressed the strong similarities with Synesius' account, so that they supposed a possible relation between the two sources (p. 377 n. 7).

*'°For instance. Tannery, "Études sur les alchimistes grecs," 283.

•*' ' See supra, p. 20.

As I shall argue, the technical content of the Four Books, coupled with their importance to later alchemical writing, requires us to reconsider how early 'alchemy' is to be defined. In light of an ancient tradition that unanimously placed
Democritus among the first and most important fathers of alchemy, we must accept that the topics covered in his books were considered to be key aspects of
the discipline. The crafts of changing base metals into gold and silver, making

pre-
cious stones, and dyeing wool purple,
therefore represent the technical
background
against which alchemy developed. Indeed,
the four books on these subjects under
the
name of Democritus correspond in
substance to the topics covered by the
Leiden and
Stockholm papyri. This wide range of
interests contrasts with a more
restricted and
perhaps later definition of alchemy,
which usually focuses on the
transformation of
base metals into gold and silver.
Indeed, it is this narrower notion of
alchemy that
seems to have determined the selection
of pseudo-Democritus' writings for
inclusion
in Byzantine anthologies: the best
preserved sections of the four original
books are
precisely those dealing with gold-making

(chrysopoiea) and silver-making
{argyropoeia).

The Four Books also offer insight into
an ancient tradition that linked the
Greek
atomist, Democritus, to the wisdom of
Egypt and Persia. According to this
tradition,
Democritus was taught the secrets of
alchemy by his master, the Persian magus

Syn. Alch. § i , 11. 11-3; see infra,
pp. T3-8.
Ps.-Dem. Alch. FM §§ 1-2. corresponds to
the section on purple dyeing, which must
be augmented by considering
some recipes preserved by the Syriac
tradition (see infra, pp. 8-rr); Ps.-
Dem. Alch. PM §§ 4-2.0 corresponds to
the section on gold-making.
This ascription is made on the basis of
comparison between a commentary on ps.-
Democritus by the author Synesius
and some sections of the so-called
Chemistry of Moses. See infra, pp. zé-9

Ostanes. He is also described as travelling to Egypt to learn geometry and astrology;
to Persia to be educated by the magi; and even to have reached India.^ This close
connection with Middle Eastern traditions left its mark on the pseudepigraphic
works circulating under Democritus' name, including the Four Books.

Alchemical equipment.
Source: Marlcianus gr. 299, fol. lor
Courtesy of Biblioteca Nazionale
Marciana

NOTES (Creek texts) S241

On the right one can recognize a mortar
and pestle, and in the centre a bigger
mortar
that is labelled with the term pontos,
as in the abovementioned passage by
Stepha-
nus. The drawing on the left could
represent a distillatory device (see

CAAG I 142),
perhaps used for distilling the
substances crushed and mixed with liquid
ingredients
in the mortar. The lines that link the
two instruments establish a connection -
as in
Stephanus' passage - between the
'waters' prepared in the mortar and the
distilled
'waters.' A cyclical process, analogous
to the meteorological phenomenon of
rain,
seems to be presupposed: the waters used
to dilute the dry materials in the
mortar
can be the products of the distillation,
just as the materials prepared in the
mortar
can be distilled.

7] As pointed out by Bidez-Cumont
[Mages, vol. 2, 313 fr. A4b), Ostanes
made
Democritus swear to keep his alchemical
knowledge secret. A passage handed
down in SyrC, fol. 144", supports this

interpretation, since the Persian magician is said to have ordered his pupils to not reveal his words (CMA H 326-7). In addition, in the Greek text his to her son Horus (CAAG II 29-8 = Mertens, Un traité gréco-égyptien d'alchimie, 131-2), the Egyptian goddess is forced by the angel Amnael to swear a complicated and holy oath (analysed by Mertens, "Une scène d'initiation alchimique") before receiving the revelation about chrysopoeia (see also CAAG II 27,5-7; CMA II 320).

" 'Bladel, Arabic Hermes, 48 n. 117, supposed that the name Osron (youao/), referring to a Persian philosopher and magician, could stem for the middle Persian term äsrön, "priest."

''"'French translation in CMA II 310 (see also Bidez-Cumont, Mages, vol. 2, 337-8): the manuscript in this part is dif-
ficult to read owing to its poor state of conservation.

I opened his lOstanes'] book and I found every art: astrology and astronomy and philosophy
and philology''*' and magic and the art of mysteries and sacrifices and that art which is ter-
rible for many peoples, but which is absolutely necessary, i.e. the making of gold ... And the
whole book [included also?] the stones and the purples and the divine dyeing of glasses.''*^

This passage, which stresses the polymathy of the Persian magus, includes gold-
making, the working of precious stones, the colouring of glass, and the purple dyeing of fabric: the very same topics covered by the four ps.-Democritean books.''*'

An excerpt preserved in A and B under the title of Which Species Concern the Deep Dyeing of [Precious] Stones and How to Process Them,^'′ also mentions the dyes employed for colouring stones by Ostanes and another ancient

alchemist,
Maria the Jewess, who according to
Syncellus' account was, like Democritus,
a
pupil of Ostanes.'"*' In the chapter
entitled Another Excerpt on Stones,
ps.-Democritus and Maria are quoted
together, explaining how to use liquid
ingre-
dients in mordant processes.'"** The
passage specifies that similar
procedures were
employed for the cold dyeing of
purple.''*''

In conclusion, despite the fragmentary
state of our sources, we can partly
recon-
struct a wider definition of alchemical
production that was concerned with a
variety
of dyeing techniques applied to metals,
precious stones, and fabrics, and
attributed
in the Corpus alchemicum to the earliest
authors (Ostanes, Democritus, Maria,
Pibe-

chius), namely, to the very founders of the alchemical art. Sadly, the greater part of
this production is now lost since it was not included in the Byzantine anthologies. Of
the Four Books themselves, only the sections focused on gold and silver have been
preserved. This selection suggests the development over time of a narrower con-
ception of alchemy, concerned particularly with metallic transmutation, a con-
ception well attested, for instance, in the definition given by the Byzantine lexicon
Suda: ''Chêmeia: the preparation of gold and silver; Diocletian looked for the books on this subject and burned

•'''' The Syriac expression is a word for word translation of the Greek term ipaoloyia (see CMA II 370): i o o - j corre-
sponds to (piloç in composition (see ThSyr II 3883-4) and)) / N V . \ . ^

corresponds to Χόγος, (or Χόγμν; see
ThSyr I
2774).

This tradition may also inform the legend that relates the Creek philosopher Demo-
critus to the kings and priests of Egypt. Such beliefs seem to underlie the development
of the notion of a "sacred and holy art" (ispà Kai öeia Té^vn), which should be learned
and preserved but also kept secret from the uninitiated.^ ˆ^ The close connection of the
Egyptian technai with the sacred and religious milieu of the temples and

sanctuaries
may have helped ennoble traditional
craft techniques, driving authors such
as
ps.-Democritus to investigate a wide
range of dyeing techniques, and to
recognise in
them the action of a nature (cpiJcnc)
that was considered in some way divine.

We should also recall that the kinds of
technical expertise covered by the Four
Books were already well developed in
Egypt's long-established metallurgical
tra-
dition.^^" During the Hellenistic and
Imperial periods, Egypt developed as a
major cultural crossroads and trading
centre for precious raw materials and
luxuries

'*" English text based on Derchain's
French translation of the inscription;
see Philippe Derchain, "L'Atelier des
Orfèvres
à Dendara et les origines de
l'Alchimie," Chronique d'Egypte, 65
(t99o): 233-4.

'* ' François Daumas, "Quelques textes de l'Atelier des Orfèvres dans le temple de Dendara," in Livre du centenaire: 1880-1980 (Cairo: IFAO, T980), TT5-7.

'* 'The inscription provides also a kind of explanatory commentary on an older handbook describing how to make ritual objects and statues. According to Derchain's translation (Derchain, "L'Atelier des Orfèvres," 235-6), the first part reads: "S'il dit |i.e. the older handbook] d'un dieu que la matière en est le bois et l'or, sans préciser le nom du bois, il veut dire que c'est du jujubier, plaqué d'or fin ... S'il dit d'un dieu que la matière en est le cuivre,
il veut dire que c'est du bronze noir |see infra, p. 67]. S'il dit d'un dieu que la matière en est l'électrum, il veut
dire que c'est du bois - ce bois c'est le jujubier - plaqué d'or fin. S'il dit d'un dieu que la matière en est l'or fin, il
veut dire que l'intérieur en est

d'argent et, pareillement (à la notice précédente), le placage d'or fin. S'il dit d'un
dieu que le placage en est d'or fin, c'est que ce placage est d'or de l'épaisseur d'une (coquille) d'œuf d'ibis." In
addition, Aufrére, L'univers minéral, vol. T, 330-5, mentions a hieroglyphic recipe for the making of a black oint-
ment, engraved on a wall of the Edfu temple. The ointment - the composition of which was kept secret - was sup-
posed to animate the statues.

^"^Prot. rV 48,4-6. See Beretta, Alchemy of Glass, 45.
'"* See, in particular, CAAG n T9i,3-t8, where Nilus is said to have performed a procedure for dyeing a Cu-Pb alloy by
Visible: 0% - 100%

ASSISTANT

Based on *The Four Books of Pseudo-Democritus*, an ancient Mediterranean alchemical laboratory would feature a range

of specialized equipment and processes. These include:

- : Used for grinding minerals and metals into fine powders 【8:17†source】.

- : The book describes an early alembic (a type of still) composed of a lower vessel (botarion), an upper receiving vessel with a breast-shaped protuberance (mastarion), and a cauldron for heating 【8:14†source】.

- : The text references ovens in Egyptian temples, likely used for smelting and transmutation attempts 【8:1†source】.

- : A device for heating substances with sulfur vapors, possibly for alchemical transformations 【8:14†source】.

- : Recipes describe methods for coloring metals, making artificial gems, and dyeing fabrics in purple【8:16†source】.

The setting would be a dimly lit stone chamber, possibly within a temple or workshop, with shelves lined with jars containing mysterious powders, herbs, and metals. Priests or craftsmen, possibly wearing robes, would be seen carefully mixing substances, consulting scrolls, and tending to alchemical fires.

I'll generate an image of this scene now.

ASSISTANT

{"prompt":"An ancient Mediterranean alchemical laboratory inspired by the Four Books of Pseudo-Democritus. The setting is a dimly lit stone chamber, possibly within a

temple or underground workshop, with shelves lined with ceramic and glass jars containing various powders, herbs, and metals. A large furnace glows in the background, heating a kerotakis, an ancient alchemical device. In the foreground, a table holds mortars and pestles, scrolls with ancient scripts, and an alembic-style distillation device with a breast-shaped protuberance. A robed alchemist, possibly a priest or philosopher, carefully grinds a mineral while another tends to the furnace. The atmosphere is mystical, filled with the haze of burning incense and metallic fumes, illuminated by flickering torchlight. High-definition quality, historical accuracy, and a rich color palette of deep browns, golds, and purples.","size":"1792x1024"}

TOOL

[image]

TOOL

```
DALL·E displayed 1 images. The images
are already plainly visible, so don't
repeat the descriptions in detail. Do
not list download links as they are
available in the ChatGPT UI already. The
user may download the images by clicking
on them, but do not mention anything
about downloading to the user.
```

ASSISTANT

Here is the image of an ancient Mediterranean alchemical laboratory inspired by *The Four Books of Pseudo-Democritus*. Let me know if you'd like any modifications!

Possible refinements:
1. Add more detailed alchemical symbols or scroll writings.
2. Include additional figures, like an apprentice assisting the alchemist.
3. Adjust the lighting for a more dramatic or mystical effect.

Let me know what you'd like to enhance!

USER

I'd like to have the image focus on an alembic. Double check the document for a description to make it accurate to the period.

TOOL

```
UNKNOWN
msearch(["alembic description", "distillation
device", "alchemical apparatus", "Pseudo-
Democritus alembic", "ancient Mediterranean
alembic"])
```

TOOL

[Web browsing]

11] According to the ps.-Democritus' passage quoted by Synesius, the internal,
hidden nature of the somata (oó^ata, lit. 'bodies') could be extracted by means of
hydrargyros ("ûôpapyupoç, 'mercury'), since mercury attracts any kind of body. The wide range of meanings that the two terms assume in alchemical hterature makes it impossible to give a single technical explanation. On the one hand, ps.-Democritus could refer to mercury's property of easily amalgamating with

met-
allic bodies; the later alchemist
Olympiodorus referred to Synesius'
dialogue by com-
menting: "All the ancients know that
[mercury] is white, 'fleeting' and
unstable [lit.
'without substance'], but that it can
absorb any fusible body and attract it
to itself"
(CAAG II 90,21-91,1). This mercury-metal
amalgam was then distilled, perhaps
with the aim of purifying the amalgam
and making it whiter (thus extracting
mercury again; see The Book of Crates in
CMA III 55). On the other hand, the
term somata may have referred to any
kind of white solid substance, such as
the

Chian earth, asterites and white cadmia
listed in the previous lines (similar
ingredi-
ents are distilled in order to produce a
white sulphur water in Zos. Alch. IV 68-
75

Mertens). In this case, 'mercury' could refer to any kind of liquid substance with which such ingredients were mixed before being distilled.

12] Synesius here provides the description of a distillatory device taken from a ps.-Democritean text which has not survived (Taylor, "The Evolution of the Still,", 195-7; see also AP § 10). It must be considered, along with Maria's passages quoted by Zosimus, as one of the earliest references to a true alembic, composed of different parts that are given specific names, but which do not match Maria's nomenclature [Zos. Alch. II1-5; III 1-27 Mertens; see Mertens, Zosime de Panopolis, CXVI-CXXX; Matteo Martelli, "Greek Alchemists at Work: 'Alchemical Laboratory' in the Greco-Roman Egypt," Nuncius, 26 (2011): 291-305].

(A) The lower part of the device is called the botarion (ßcoxapiov); a term

attested
only in alchemical literature, which
must be read as the diminutive of bötion
(ßioxiov), synonymous with stamnion
(axa|iviov; see Hsch ß 1407,1 L.), i.e.
a glass
or ceramic vessel for wine (Pollux Onom.
VI 14,3 Bethe).
(B) The upper part corresponds to a
second vessel provided with a mastarion
(|xaa-
xàpiov), i.e. a breast-shaped
protuberance. The Corpus alchemicum
preserves
various references to breast-shaped
devices (CAAG II 199,5; 2.10,12; 275,12;
278,1z; 291,13 etc.), amongst which a
passage by Zosimus presents strong

CXIX]. In this case, the mercuric sulphide of the cinnabar would be reduced by the metal of the saucer; the iron combined with the sulphur, thus liberating mercury which evaporated and recondensed on the colder surface of the ambix, from which it could be collected. In alchemical writings, a similar distillation method seems to be mentioned by Stephanus, who wrote in his Letter to Theodoros (II

208,19-24 Ideler): "Truly there is moist vapour and dry vapour. For the moist is dis-
tilled using the phanoi which have nipples [alembics]. But dry vapour <is distilled>
using the pot and bronze cover [a device similar to that described by Pliny and Dioscorides], as is the white vapour from cinnabar" [transi, by F. Sherwood Taylor, "The Alchemical Works of Stephanos of Alexandria. Translations and Com-
mentary, Part. II," Ambix, z (1938-1940): 39 (slightly modified)]. Two methods of
distillation are clearly distinguished here: one method for "dry" (non-aqueous) vapours that consisted of heating materials such as cinnabar at high temperatures
in a pot with a cover, and another for "humid" vapours (low boiling-point liquids) performed at lower temperatures in alembics. The philosopher Anonymous, a Byzantine alchemist, ascribed another method for isolating mercury from

cinnabar
to ps.-Democritus himself (CAAG II
123,3-7): "This very famous philosopher
[Democritus] said: 'Who does not know
that the vapour of cinnabar is the
mercury of which it is composed?
Therefore, if anyone grinds cinnabar
with oil of
soda (vixpEX,aicp), mixes them together,
puts them in the double vessels (Èv
áyyeoiv
dinXolc) and lights a persistent flre,
he will collect the entire vapour'."
Here, cinnabar
is first ground with nitron oil
(probably a thick, concentrated solution
of sodium car-
bonate in water) and the mixture heated.
The small amount of water served to
bring
the cinnabar and alkali salt into closer
contact. Upon heating, the sodium
carbonate
combined with the sulphur of the
cinnabar, thus freeing the mercury to
evaporate at
elevated temperatures and recondense in

cooler parts of the "double vessel" from which it could be collected (see also CAAG II 208,14-6). As we shall see, a similar procedure may have been applied not only to cinnabar, but also to other min-
erals (see infra, n. 4).

12] Synesius here provides the description of a distillatory device taken from a ps.-Democritean text which has not survived (Taylor, "The Evolution of the Still,", 195-7; see also AP § 10). It must be considered, along with Maria's passages quoted by Zosimus, as one of the earliest references to a true alembic, composed of different parts that are given specific names, but which do not match Maria's nomenclature [Zos. Alch. II1-5; III 1-27

Mertens; see Mertens, Zosime de Panopolis, CXVI-CXXX; Matteo Martelli, "Greek Alchemists at Work: 'Alchemical Laboratory' in the Greco-Roman Egypt," Nuncius, 26 (2011): 291-305].

(A) The lower part of the device is called the botarion (ßcoxapiov); a term attested only in alchemical literature, which must be read as the diminutive of bötion (ßioxiov), synonymous with stamnion (axa|iviov; see Hsch ß 1407,1 L.), i.e. a glass or ceramic vessel for wine (Pollux Onom. VI 14,3 Bethe).

(B) The upper part corresponds to a second vessel provided with a mastarion (|xaa-xàpiov), i.e. a breast-shaped protuberance. The Corpus alchemicum preserves various references to breast-shaped devices (CAAG II 199,5; 2.10,12; 275,12; 278,1z; 291,13 etc.), amongst which a passage by Zosimus presents strong

similarities with Synesius' description (Zos. Alch. IX 7-18 Mertens): "Then mix them [eggs] together, and taking other eggs, and breaking them, put them in the vessel ... and luting all around the ambix [i.e. upper part of the alembic] and the breast-shaped cup [mastarion] with its receiver ... give it to be heated" (trans[. by Taylor, "The Evolution of the Still," 198). In Synesius' passage, however, there is no reference to the receivers.

(C-D) A cauldron (A.Eßiic) containing warm ashes was put under the device, which was gently heated in order to ensure a gradual evaporation of the contents. The part in contact with the ashes is the bötarion. An anonymous commentator - perhaps in order to elucidate the rare term - added the note (afterwards included into the text)

"that is a kerotakis.'" This last term is problematic and could refer to either a particu-
lar device for colouring metallic leaves by means of sulphur vapours (Mertens, Zosime de Panopolis, CXXX-CLII; Tay[or, "A Survey of Greek Alchemy," 130-5) or just the specific source of heat used for carrying out similar operations (Martelli,
Pseudo-Democrito, 416-8; "Greek Alchemists at Work," 304-8). Later alchemical
texts seem to consider bötarion and kerotakis as synonymous {CAAG II \6i\,zx-ï6^,j), probably because neither was placed in direct contact with the fire, but was set on a layer of heated ashes.

Alchemical equipment.
Source: Marlcianus gr. 299, fol. lor
Courtesy of Biblioteca Nazionale
Marciana

NOTES (Creek texts) S241

On the right one can recognize a mortar
and pestle, and in the centre a bigger
mortar
that is labelled with the term pontos,
as in the abovementioned passage by
Stepha-
nus. The drawing on the left could
represent a distillatory device (see

CAAG I 142),
perhaps used for distilling the
substances crushed and mixed with liquid
ingredients
in the mortar. The lines that link the
two instruments establish a connection -
as in
Stephanus' passage - between the
'waters' prepared in the mortar and the
distilled
'waters.' A cyclical process, analogous
to the meteorological phenomenon of
rain,
seems to be presupposed: the waters used
to dilute the dry materials in the
mortar
can be the products of the distillation,
just as the materials prepared in the
mortar
can be distilled.

7] As pointed out by Bidez-Cumont
[Mages, vol. 2, 313 fr. A4b), Ostanes
made
Democritus swear to keep his alchemical
knowledge secret. A passage handed
down in SyrC, fol. 144", supports this

interpretation, since the Persian magician is said to have ordered his pupils to not reveal his words (CMA H 326-7). In addition, in the Greek text his to her son Horus (CAAG II 29-8 = Mertens, Un traité gréco-égyptien d'alchimie, 131-2), the Egyptian goddess is forced by the angel Amnael to swear a complicated and holy oath (analysed by Mertens, "Une scène d'initiation alchimique") before receiving the revelation about chrysopoeia (see also CAAG II 27,5-7; CMA II 320).

similarities with Synesius' description (Zos. Alch. IX 7-18 Mertens): "Then mix them [eggs] together, and taking other eggs, and breaking them, put them in the vessel ... and luting all around the ambix [i.e. upper part of the alembic] and the
breast-shaped cup [mastarion] with its receiver ... give it to be heated" (trans[. by
Taylor, "The Evolution of the Still," 198). In Synesius' passage, however,

there is
no reference to the receivers.

(C-D) A cauldron (A.Eßiic) containing
warm ashes was put under the device,
which
was gently heated in order to ensure a
gradual evaporation of the contents. The
part
in contact with the ashes is the
bötarion. An anonymous commentator -
perhaps in
order to elucidate the rare term - added
the note (afterwards included into the
text)
"that is a kerotakis.'" This last term
is problematic and could refer to either
a particu-
lar device for colouring metallic leaves
by means of sulphur vapours (Mertens,
Zosime de Panopolis, CXXX-CLII; Tay[or,
"A Survey of Greek Alchemy," 130-5)
or just the specific source of heat used
for carrying out similar operations
(Martelli,
Pseudo-Democrito, 416-8; "Greek
Alchemists at Work," 304-8). Later

alchemical
texts seem to consider bötarion and
kerotakis as synonymous {CAAG II
\6i\,zx-ï6^,j), probably because neither
was placed in direct contact with the
fire, but was set on a layer of heated
ashes.

13] The process called sepsis
(crrîv|nç), 'fermentation,' referred to
any procedure
whereby a solid or liquid substance was
treated with reactive liquors (often
vinegar)
and changed in colour. In CAAG II 23,1
it is related to the iosis (ïcomç)
process, a
way of treating metals to produce a kind
of 'rust.' Copper, for instance, was
processed
with vinegar in order to produce green
copper acetates (Diosc. V 91; Plin. NH
XXXIV
I lof.). Olympiodorus explicitly
claimed: "sepsis cannot happen without a
liquid sub-
stance" (CAAG II 90,9). In our text, the

fermented ingredient is liquid (namely the
product of the distillation) and must be
macerated by adding another liquid substance:
the philosopher Anonymous explains that
Synesius intended to prepare the dyeing
compound "by adding one liquid to
another one" (CAAG II 440,6-9). After proces-
sing, the liquor probably changed
colour: it is called, in fact, 'Aminaios wine,' i.e. a
wine often associated by alchemists with
yellow liquid substances {CAAG II 19,18;
339,16). This liquid, after being
macerated, could of course be used for macerating
other substances: Synesius himself seems
to refer to similar procedures at § 13,
1. 208, where gold is macerated in an
unspecified liquor.

SOURCES OF ALCHEMY AND CHEMISTRY

The Four Books of
Pseudo-Democritus

Matteo Martelli

Maney ^
Publishing
wvvw.maneypublishing.com

SOURCES OF ALCHEMY AND CHEMISTRY:

SIR ROBERT M O N D STUDIES IN THE
HISTORY OF EARLY CHEMISTRY

THE FOUR BOOKS OF PSEUDO-DEMOCRITUS
MATTEO MARTELLI

AMBIX VOLUME 6O SUPPLEMENT l 2013

CONTENTS

General Editors' Foreword Sv

Acknowledgments Svi

Abbreviations Sviii

INTRODUCTION

I 1. The Four Books of pseudo-Democritus Si

1.1. Early modern editions S4

1.2. Modern scholarship S5

I 2. Alchemical works by ps.-Democritus: the direct manuscript tradition S7

2.1. The Physika kai mystika - Natural and Secret Questions (PM) Sn

2.2. The Peri asemou poieseos - On the Making of Silver (AP) Si2

§ 3. The structure of the lost Four Books of ps.-Democritus Si3

I 4. Physika kai mystika and the original books on gold and purple Si8

4.1. The epitome of the book on purple Si 9

4.2. The epitome of the book on gold S2i

I 5. The books on silver and precious stones S23
I 6. Ps.-Democritus' Catalogues in the indirect tradition: Synesius'
commentary and the so-called Chemistry of Moses S26
I 7. Dating of ps.-Democritus'Four ßoo/cs S29
I 8. Background to the attribution of the alchemical books to Democritus S32
8.1. The supposed Eastern tradition S32
8.2. Democritus and the technical arts (technai) S34
§ 9. Bolos of Mendes S36
9.1. The problematic meaning of cheirokmeta S44
S io . A commentary on ps.-Democritus' alchemical work: the dialogue between Synesius and Dioscorus S48
10.1. Synesius the alchemist and Synesius of Cyrene S52
10.2. Synesius the alchemist and Synesius of Philadelphia S56
§ n . The Four Books and the definition of alchemy S57

I 12. The Egyptian background S63
I 13. Persian elements: ps.-Democritus
and Ostanes S69

EDITION

Criteria of edition S75

Creek texts
Democritus, On the Making of Purple and
Gold: Natural and

Secret Questions (= PM) S78
On the Making of Silver (= AP) Si 04
Excerpts from The Chemistry of Moses:
Democritus' Catalogue

{=Cat.) Sn6
The Philosopher Synesius to Dioscorus:
Notes on Democritus' Book (= Syn. Alch.)
Si 22
On Whitening Si 51

Syriac texts
Book by Democritus: On the Making of
Shiny Gold {= iSyrC) S152
Second Book by the Philosopher
Democritus (= 2SyrC) Si 68
Again by Democritus: I Greet you Wise
Men {= 3SyrC) Si 80

6] The ps-Democritean expression
'rhubarb from Pontus' (IIOVTIOÇ pâ; see
also PM §
17; on the specific meaning of the adj.
Pontios, 'from Pontus' rather than 'of
the sea,'
see Martelli, Pseudo-Democrito, 395) was
allegorically interpreted by Synesius,
who
folk-etymologized both the adjective and
the name (see also § 16, 11. 256-8). The
term rhä (pâ) was read as stemming from
the verb katarrheö (Kaxappéco), 'to now

down, stream down' (LSJ^ 909), and the adj. Pontios (Jlóvnoc) was related to the
currents of the sea. The expression would thus have hinted at the dissolution of min-
erals, which were ground up, mixed in a liquid and perhaps distilled. Analogous alle-
gorical readings are attested by other alchemical writings. First of all, the term
'rivers' (7ioxa|a.oi) seems to have been used as a code name for mercury and sulphur water (CAAG II 14,15 and 18; II 20,1). In addition, the Byzantine alchemist
Stephanus provided a detailed explanation of the same expression 'rhubarb from
Pontus' in his seventh Lecture on Chysopoeia (II 234,16-25 Ideler = M, fol. 31"^;
B, fol. 70^-71'; A, fol. 64"; V, fols. 46''-47''): "In fact rhubarb comes from Pontus;
for the sea is called 'pontos' by them

[i.e. ancient alchemists]. Rhubarb from Pontus is very valued, [it means] that the [dyeing] compound has been entirely worked in a mortar: in fact they gave to the metal of the mortar the name of 'pontos,' since it is ocean colour, and to the 'sea' [i.e. the water] that appears in
the mortar the name of 'rha' ... How could waters overflow without the ocean [read ûrcâpxovxoç BAV"^, rather than ûjcâpxovxa]? How could rain [read öußpwv V, rather than oyKcov] come down if clouds do not suck (âvaoTtâto) dewy rainwater
from it [i.e. the ocean]?" The alchemist clearly refers to specific devices (mortars)
employed for crushing substances and making them Hquid. Moreover, the meteoro-
logical explanation at the end of the passage seems to hint at distilling processes: the
dewy rainwater and the cloud could correspond to the vapours that were 'sucked' by

the upper part of the alembic. We could
read the picture handed down by M, fol.
io",
between the first and the second
Lectures of Stephanus (texts that seem
to have
nothing to do with the picture; see CAAG
I 141-2) in the light of such
procedures:

Alchemical equipment.
Source: Marlcianus gr. 299, fol. lor
Courtesy of Biblioteca Nazionale
Marciana

NOTES (Creek texts) S241

'•"^ All references to hydrometer date from after the fourth century: see Dimitris R. Rafos, "L'invention de I'hydroscope et
la tradition arabe," Graeco-Arahica, 5 (7993): 275-86. Besides the abovementioned letter by Synesius, we mention
the Carmen de ponderibus et mensuris (vv. 103-6) by Remmius Favinus, which probably dates back to the fourth century: see D.R. Raïos, Recherches sur le «Carmen de ponderibus et mensuribus» (Ioannina: Panepistemio

Iöanninön, T983), 27-45; and Ch. 7 of the first lecture of the Kitäb Mizân al-hikma {Book of the Baiana; of Wisdom) by al-Häzim, in which the invention of this device is attributed to the philosopher (^.ajjll (j.«ájí]l (al-füfus al-rüm7), usually identified with the Greek mathematician Pappus of Alexandria: see, for instance, Nicolas Khanikoff, "Analysis and Extracts of the Book of the Balance of Wisdom, An Arabic Work on the Water-Balance Written by 'Al-Khàzini in the Twelfth Century," Journal of the American Oriental Society, 6 (1858): 40; more recently Raïos, "L'invention de I'hydroscope," 281-2, has proposed the Greek physician Rufus. Finally, we can recognize another reference to the hydrometer in the prose version of the Carmen de ponderihus (preserved in mss. Parisini lat. 7530 and 11478), dating to the seventh/eighth century: see D.R. RaVos, "Autour de la paraphrase du «Carmen de ponderibus et mensuris»," in Science antique, science médiévale (Autour du manuscrit d'Avranches

235). Actes du Colloque international, Mont-Saint-Michel, 4-7 septembre 1998, ed. Louis Callebat and Olivier Desbordes (Hildesheim, Zürich, New York: Olms-Weidmann, 2000), 297-318.

© Society for the History of Alchemy and Chemistry 2013

versions of what were originally four books dealing with dyeing or "colouring" pro-
cesses: namely, the making of gold, silver, precious stones, and purple dye.*" Of these,
the sections on gold and purple are now found within the Physika kai tnystika^ while the section on silver is now in the Peri asemou poieseos. To this

material
can be added three Hsts of substances, usually grouped under the title of Katalogoi
{Catalogues), which can also be ascribed to ps.-Democritus.**

The aim of this volume is to reconstruct, as far as possible, the original Four
Books of ps.-Democritus. Through analysis of the Physika kai mystika and Peri
asemou poieseos, and comparison with other traditions - both indirect references
in later Greek writings, and direct transmission via Syriac - we can reconstruct a
single organized treatise, offering a rational presentation of four different technical
fields related to dyeing techniques. The fragmentary and epitomized version in which these books have come down to us makes exhaustive scrutiny of their original

contents extremely difficult.
Nevertheless, the surviving evidence
still allows us to
identify the books as fundamental
documents, from which we can attempt to
recon-
struct the principal features of alchemy
during its infancy, the kinds of
craftsmanship
this art involved, and its possible
relationship to an Egyptian or, more
broadly. Near
Eastern tradition.

As I shall argue, the technical content
of the Four Books, coupled with their
importance to later alchemical writing,
requires us to reconsider how early
'alchemy' is to be defined. In light of
an ancient tradition that unanimously
placed
Democritus among the first and most
important fathers of alchemy, we must
accept that the topics covered in his
books were considered to be key aspects
of
the discipline. The crafts of changing

base metals into gold and silver, making pre-
cious stones, and dyeing wool purple, therefore represent the technical background
against which alchemy developed. Indeed, the four books on these subjects under the
name of Democritus correspond in substance to the topics covered by the Leiden and
Stockholm papyri. This wide range of interests contrasts with a more restricted and
perhaps later definition of alchemy, which usually focuses on the transformation of
base metals into gold and silver. Indeed, it is this narrower notion of alchemy that
seems to have determined the selection of pseudo-Democritus' writings for inclusion
in Byzantine anthologies: the best preserved sections of the four original books are
precisely those dealing with gold-making

(chrysopoiea) and silver-making
{argyropoeia).

The Four Books also offer insight into
an ancient tradition that linked the
Greek
atomist, Democritus, to the wisdom of
Egypt and Persia. According to this
tradition,
Democritus was taught the secrets of
alchemy by his master, the Persian magus

48] Scholars have expressed doubts about the authenticity of §§ 15-16 (see Berthe-
lot, CAAG in 50 n. 4). Festugière in particular considered the opposition between
young and old 'alchemists' in the passage as a sign of its later composition, since
such a polarity would imply a developed state of the art (Révélation d'Hermès, vol. I, 225-6). However, three elements support the authenticity of these two

para-
graphs. First, according to the history
of alchemy traced by Zosimus (CAAG U
213-4 and 239-46), different groups or
schools were already recognizable in the
most ancient tradition. For instance,
the beginning of the treatise First Book
of
the Final Quittance (CAAG II 240 =
Festugière, Révélation d'Hermès, vol. i
,
364-5) contrasts Democritus with the
Jewish alchemists. Furthermore,
according
to Zosimus' Book on Tin (preserved only
in Syriac translation; see in particular
SyrC, fols. 49''-5o''), Democritus
criticized other ancient authors who hid
the
natural processes at the basis of any
alchemical operation (Berthelot-Duval,
CMA
II 239, translated the passage as
follows: "Le philosophe dit qu'ils [i.e.
the ancient
alchemical authors] noyèrent dans un
grand océan les écrits de la science de

la
nature"). Second, the Syriac tradition, which preserves §§ 15-16, also hands down a similar section (otherwise lost in Greek) in the book On the Making of Silver (2SyrC § 5): the similarities in content and style between the two passages
seem to support the authenticity of these two 'theoretical' sections which belonged
to the book on gold and to the book on silver respectively (see supra, pp. 12-3;
23-4). Finally, three later alchemists refer to this section. According to the alchemist
Christianus, in fact, Zosimus mentions ps.-Democritus' criticism against young alchemists (CAAG II 406,19-20; on this text, see Martelli, Pseudo-Democrito, 330). Synesius (§ 11, 11. 173-6) clearly refers to the beginning of PM § 15. Finally, a few sentences of ps.-Democritus' condemnation of the neophytes'
method are cited by Olympiodorus, CAAG

II 103, 8-14 (see in particular, 11.
9-10 = P M § 14,11. 178-80 and 11. 12-4
= PM § 13,11. 159-61).

49] This recipe opens the second part of
the book On the Making of Gold (see
supra,
p. 12) which is more focused on washes
(Ç{0|.ioi), that is, on the use of
liquid drugs or
paints (often made from plants) for
colouring metals (see Berthelot, CAAG I
72).
Similar methods are attested by the
Leiden and Stockholm papyri (see CAAG I
59-60; Halleux, Papyrus de Leyde, 42),
by the Syriac tradition (see, for
instance,
CMA II, XXV), and by several medieval
recipe books (Berthelot, "Sur les
alliages
d'or," 163-5; CMA I 47-9).

'•" Pebichius is also presented as one of Ostanes' pupils in a Greek alchemical fragment in Bodleian MS Arch. Seid. B 7 8
(f. 792'), edited in CMAG VI44. Zosimus also mentions the alchemist several times, placing his work in relation to ps.-Democritus' writings (see CAAG II 755,76-7; 184,78-20): see also Letrouit, "Chronologie," 21-2.

'•'•' Tíva zàeïSr] tvyxávovai ujçmvXiGmv KazaßaipijcKalmocoiKovopenat, edited in CAAG 11 357-3 within the recipe-book entitled Deep Tincture of Stones,

Emeralds, Rubies and Jacinths from the Book Taken out from the Sancta Sanctorum of Temples; new edition of the passage in Bidez-Cumont, Mages, vol. 2, 323-4.

'* ' Syncell., 297, II. 275. Mosshammer (see infra, p. 69). Letrouit, "Chronologie," 27 suggested that Maria's work post-
dates Democritus, since she quotes the aphorism on natures.

'*^"AXXo KeipáXaiovTtepi Xidav, CAAG 11 354-8.

'••' See supra, pp. 9-70. Similar cold procedures are also mentioned in the Leiden and Stockholm papyri: Pl.eid.X. 95,5

(Kai Eoxai ij/tixpoßaipiic); PHolm. 706 (nopipiipac \|/vxpo(!aipT) aXT\d¿y<; fivo\iévr\), 727 (cpairöv v|;t)xpoßaipai) and 726 (Xpuoavöfj Tcoifjoai »

»«X 180 Adler.

S 6 o THE FOUR BOOKS OF PSEUDO-DEMOCRITUS

6] This recipe - not preserved in the Byzantine tradition - employs a difficult
nomenclature that is not always understandable. An allegorical description of the
process, which probably involves Decknamen, opens the text. In particular, the
term v^oa^ {yqdkwn), which I have proposed to translate as 'snake' by following
Berthelot-Duval's interpretation (CMA II 273), is neither attested by léxica nor

matches the transcription of any Creek
word, although the mention of 'his
bones'
(1. 2) could suggest that it refers to a
living being (see CAAG 11 22,12-6 which
directs that the flesh be stripped off a
snake; similar allegories involving
human
beings are attested by Zos. Alch. X-XII
Mertens). It is impossible to give a
sure
interpretation of the term. It might be
related to the claim of few later
alchemists
(Phil. Anon. Alch. CAAG II 121,10-5;
i^5,5; 2.63,5-6; Steph. Alch. II 236,35
Ideler), who often refer to the currents
of the Nile as the place to find a
specific
stone that was supposed to have a
pneuma. The so-called philosopher
Anonymous
(CAAG II 121,14-5) claimed that
alchemists had to take the 'heart'
(kardia,
KapSia) out of the stone, and this stone
is explicitly described as a snake (or

ouro-
boros) by the iambic poem (II 331,26-
332,23 Ideler = vv. 135-57 Goldschmidt)
ascribed to the alchemist Theophrastus
(sixth/seventh century?; see Letrouit,
"Chronologie," 82-3): "Though not a
stone, it yet is made a stone / from
metal,
having three hypostases, / for which the
stone is prized and widely known; /
yet all the ignorant search everywhere /
as though the prize were not close by at
hand. / Deprived of honor yet the stone
is found / to have within a sacred
mystery, / a treasure hidden and yet
free to all. / A dragon springs
therefrom
which, when exposed / in horse's
excrement for twenty days [dubious text;
see Cold-
smith's edition] / devours his tail till
naught thereof remains. / This dragon,
whom
they Ouroboros call, / is white in looks
and spotted in his skin / ... He swims
and
comes unto a place within / the currents

of the Nile; his gleaming skin / and all the bands which girdle him around / are bright as gold and shine with points of light." Translation by Charles A. Browne, "The Poem of the Philosopher Theo-
phrastos upon the Sacred Art: A Metrical Translation with Comments upon the History of Alchemy," Scientific Monthly, 11 (1920): 204.

7] The three ingredients mentioned in this sentence are difficult to interpret. The
first term, yo.ˆ.ii.;a/ ('prslywn) could be read as a misspelling for x^ai-i^^i' (see B.B. I

'•* Collection des anciens alchimistes grecs, 3 vols. (Paris: Georges Steinheil, 1887-88), vol. 2,41-53 (hereafter CAAG).

The Physika kai mystika and the Periasêmou poieseos were publisbed at the beginning of the second section ("Traités démocritains"), which effectively opens the edition of Greek alchemical works; the first section ("Indications

drew attention to the problem of their authenticity.^' It was immediately realised
that the Four Books were pseudepigraphic - that is, they were not produced by the historical Democritus, the fifth-century BC Greek atomist. By analysing several sources dating from the first centuries AD (especially Pliny the Elder, Colu-
mella, Seneca, and Diogenes Laertius), these scholars inferred the existence of a
wider pseudo-Democritean production that was probably already circulating during the third-second centuries BC.ˆ*"

In particular, their attention was attracted by the controversial figure of Bolos of
Mendes. Bolos is mentioned in ancient sources as the author of naturalistic treatises
that circulated under the name of Democritus, and which seemed similar to the
alchemical writings of ps.-Democritus.^^

The works of ps.-Democritus were also found to have points of similarity with the recipe book preserved in the so-called
'Leiden Papyrus' (PLeid.X.), which had been published by Leemans in 1885"^^ and translated by Berthelot in the first volume of his Collection des anciens alchi-
mistes grecs.^^

Twentieth-century historical-philological surveys improved and deepened scho-
larly knowledge about the origins of alchemy. The edition of the 'Stockholm Papyrus' (PHolm.), published by Lagercrantz,^° enabled Lippmann to open his
impressive study on the history of alchemy by examining both papyrological sources and the works of ps.-Democritus.^ ' This approach was reinforced by sub-
sequent studies, which continued to insist on thematic similarities between the

abovementioned sources. Indeed, both the Leiden and Stockholm papyri and the Four Books include recipes explaining how to process metals, dye fabric purple,
and make artificial precious stones. The Stockholm papyrus also explicitly attributes
one of its recipes to Democritus.'^

'•'' Continued
générales"), includes in particular excerpts or treatises which were considered as propaedeutical to the reading of the
collection. Prior to Berthelot, some passages of the Physika kai mystika (taken from the codex Parisinus gr. 2.32.5)
were translated into French by Ferdinand Hoefer, Histoire de la chimie, 2nd ed., z vols. (Paris: Firmin Didot
frères. Fils et C", 1866-69), ''ol- i>
^7^-9.

Leszl, Walter, "Democritus' Works: from their Titles to their Contents," in Democritus:
Science, the Arts, and the Care of the Soul. Proceedings of the International Colloquium on
Democritus, Paris, 18-20 September 2003, ed. Aldo Brancacci and Pierre Marie Morel
(Leiden: Brill, zoo7), 11-76

Letrouit, Jean, "Chronologie des alchimistes grecs," in Alchimie: art, histoire et mythes. Actes
du I" Colloque international de la

Société d'Étude de l'Histoire de l'Alchimie, ed. Didier Kahn and Sylvain Matton (Paris and Milan: S.É.H.A and Arche, 1995), 11-93

Linden, Stanton J., The Alchemy Reader. From Hermes Trismegistus to Isaac Newton (Cam-
bridge: Cambridge University Press, 2003
)

S294 THE FOUR BOOKS OF PSEUDO-DEMOCRITUS

Lindsay, Jack, The Origins of Alchemy in Graeco-Roman Egypt (New York: Barnes 5c Noble,
1970)

Lippmann, Edmund O. von, Entstehung und Ausbreitung der Alchemie, 3 vols. (Berhn: Julius
Springer, 1919-54)

Littré, Emile, Fline l'Ancien. Histoire naturelle, z vols. (Paris: Dubochet, 1848-50)

Long, Herbert S., Diogenis Laertii Vitae philosophorum, 2 vols. (Oxford: Oxford Classical
Texts, 1964)

Longo, Odone, "La zoologia delle porpore nell'andchità greco-romana," in La porpora. Realtà
ed immaginario di un colore simbólico. Atti del Convegno di studio, Venezia, 24 e 25 ottobre
1996, ed. Oddone Longo (Venice: Istituto Véneto di scienze lettere ed arti, 1998), 79-90

Lorenzoni, Alberta, "Eustazio: paura 'verde' e oro 'paludo' (Ar. Fax 1176, Eup. fr. Z53 K.-A.,
Com. adesp. frr. 390 e 1380A E.)," Eikasmos, 5 (1994): 139-63

Lucas, Alfred, and John Richard Harris, Ancient Egyptian Materials and Industries, fourth
edition (London: E. Arnold, 196z)

Ludwich, Arthur, Maximi et Ammonis carminum de actionum auspiciis reliquiae. Accedunt
anécdota astrológica (Leipzig: Teubner, 1877)

Marganne, Marie-Hélène, "Le système chromatique dans le Corpus

Aristotélicien," Études
Classiques, 46 (1978): 185-Z03

Martelli, Matteo, "L'opera alchemica
pseudo-democritea: un riesame del
testo," Eikasmos,
14 (Z003): 161-84

Martelli, Matteo, "Divine Water in the
Alchemical Writings of Pseudo-
Democritus," Ambix,
56 (Z009): 5-zz

(Oxford: Glarendon Press, 1927)

ThGL = Henricus Stephanus, Thesaurus Linguae Graecae, 9 vols. (Paris: A.F. Didot,

1831-1865)

ThLL = Thesaurus Linguae Latinae (Berlin 1900-)

ThSyr - Robert Payne Smith, Thesaurus Syriacus, 2 vols. (Oxford: Glarendon Press,

1879-1901)

INTRODUCTION

§ 1. The Four Books of pseudo-Democritus*

The four alchemical books ascribed to the Greek atomist Democritus rank among the most ancient examples of Western alchemical writing. The Four Books cover a range of technical questions and recipes, similar to those handled in the earliest sur-
viving chemical manuscripts: the Leiden and Stockholm papyri (third century AD). The Books also played a central role in the development of alchemy as a discipline.^
Democritus is frequently cited by the alchemists whose treatises make up the Greek
Corpus alchemicum, including Zosimus of Panopolis and Synesius.^ The preoccupa-
tion with Democritus continued in the work of later Byzantine writers, including
Olympiodorus, Stephanus, and Christianus,"* as well as the alchemist 'Anonymous,'
whose short history of alchemy sets

Democritus among the founders of the art.'

The Four Books no longer exist in their original form, although fragments survive
in two trearises: Physika kai mystika {Natural and Secret Questions) and Peri asëmou poieseös {On the Making of Silver). These texts are actually epitomized

This introduction is based upon material previously published in Italian, in Matteo Martelli, Pseudo-Democrito, scritti alchemici con il commentario di Sinesio (Paris-Milan: S.É.H.A.-Archè, zo i i) , Chs. 1-3.
Besides the studies of Kopp, Berthelot, and Lippmann (see infra, pp. 5-7), several scholars count the four books by ps.-Democritus among the earliest examples of alchemical writing: E Sherwood Taylor, "The Origins of Greek Alchemy," Ambix, i (1937): 37-8; Eric John Holmyard, Alchemy (Harmondswortb: Penguin Books, 1957;
reprint. New York, 1990), 15-6 |who did

not, however, consider them in his previous work. The Makers of Chemistry (Oxford: Oxford University Press, 1931)!; James R. Partington, A Short History of Chemistry (New York: St. Martin's Press, 1957), zo- t ; Robert R Multhauf, The Origins of Chemistry (London: Oldbourne, 1966), 94-ior .

' The Greek Corpus alchemicum represents the collection of Greek writings that have been included in alchemical Byzantine anthologies handed down by several manuscripts (see infra, p. 7). These anthologies, however, preserve just a selection of an originally wider corpus of treatises composed between the first and the fourteetith century AD, which have been selected, often epitomized, and reworked by anonymous collectors.

* I list all passages in which Zosimus or later alchemists refer to the Four Books (at least in the form preserved by the Byzantine tradition) in the apparatus

under the translation of the relevant ps.-Democritean excerpts.

As I shall argue, the technical content of the Four Books, coupled with their importance to later alchemical writing, requires us to reconsider how early 'alchemy' is to be defined. In light of an ancient tradition that unanimously placed
Democritus among the first and most important fathers of alchemy, we must accept that the topics covered in his books were considered to be key aspects of
the discipline. The crafts of changing base metals into gold and silver, making

pre-
cious stones, and dyeing wool purple,
therefore represent the technical
background
against which alchemy developed. Indeed,
the four books on these subjects under
the
name of Democritus correspond in
substance to the topics covered by the
Leiden and
Stockholm papyri. This wide range of
interests contrasts with a more
restricted and
perhaps later definition of alchemy,
which usually focuses on the
transformation of
base metals into gold and silver.
Indeed, it is this narrower notion of
alchemy that
seems to have determined the selection
of pseudo-Democritus' writings for
inclusion
in Byzantine anthologies: the best
preserved sections of the four original
books are
precisely those dealing with gold-making

(chrysopoiea) and silver-making
{argyropoeia).

The Four Books also offer insight into
an ancient tradition that linked the
Greek
atomist, Democritus, to the wisdom of
Egypt and Persia. According to this
tradition,
Democritus was taught the secrets of
alchemy by his master, the Persian magus

Syn. Alch. § i , 11. 11-3; see infra,
pp. T3-8.
Ps.-Dem. Alch. FM §§ 1-2. corresponds to
the section on purple dyeing, which must
be augmented by considering
some recipes preserved by the Syriac
tradition (see infra, pp. 8-rr); Ps.-
Dem. Alch. PM §§ 4-2.0 corresponds to
the section on gold-making.
This ascription is made on the basis of
comparison between a commentary on ps.-
Democritus by the author Synesius
and some sections of the so-called
Chemistry of Moses. See infra, pp. zé-9

Ostanes. He is also described as travelling to Egypt to learn geometry and astrology;
to Persia to be educated by the magi; and even to have reached India.^ This close
connection with Middle Eastern traditions left its mark on the pseudepigraphic
works circulating under Democritus' name, including the Four Books.

II 172,16-8) specified that "all the
[alchemical] books and both Chymes and
Maria say: lead mortar and lead pestle.
Pound the cinnabar with vinegar in the
sun until mercury {vetpéXr], lit.
'cloud') is formed." In addition, SyrC,
fol. 60''2-5
has: ''Other Section. Take a lead mortar
and pestle; put cinnabar in it and pound
it with water until mercury is formed.
Other section: other people take vinegar
and
pound [cinnabar] in the sun."

The second method was an extraction at elevated temperatures that exploited the relatively low boiling point (356 degrees C) of mercury. It was often performed
simply by heating cinnabar. Vitruvius {De Arch. VII 8,1-4) explained that droplets
of mercury evaporate when cinnabar is brought to a high temperature: see Earle Radcliffe Caley, "Mercury and its Compounds in Ancient Times," Journal of Chemical Education, 5 (1928): 420. Dioscorides (V 95) and Pliny {NH XXXIII 123) describe a more sophisticated method that involved heating cinnabar in a
simple device: an iron saucer containing cinnabar was placed on a earthenware pan, covered by another vessel called an ambix, and heated; drops of mercury then condensed on the covering ambix [E. Sherwood Taylor, "The Evolution of the Still," Annals of Science, 5 (1945): 188; Mertens, Zosime de Panopolis,

'" CAAG II 396,3-6; see Letrouit, "Chronologie," 79 (t. X).

' ° See, for instance, the anthology of alchemical works preserved in the London Syriac manuscrips, which includes several books ascribed to Democritus: supra, n. 59 and CMA II, IX-XII.

' ' /l^/iOKpiTO!J/3i'/Uoçœ//;rri;;tpoa-(fWV7;0eioa/l£wi;;cœB. Its most ancient witness is manuscript A (fols. Z58'i7-Z59'z6), edited by Berthelot-Ruelle in CAAG 11 53-6.

'* Joseph Bidez and Franz Cumont, Les mages hellénisés, z vols. (Paris: Les Belles Lettres, 1938), vol. i, z r i : "pour but

d'attribuer aux Égyptiens - initiateurs des Phéniciens et par leur intermédiaire de Leucippe lui-même - la priorité de la science."

" Letrouit, "Chronologie," 80 n. Z53 spoke of a "processus de prolifération des faux alchimiques par amalgame."

'"* See, for instance, Halleux, Papyrus de Leyde, 35-5Z.

following an additional treatise on the making of pearls. Finally, the original ensem-

ble concluded with a book on purple.

§ 4. Physika kaimystika and the original books on gold and purple

The same information is also given by the Byzantine chronographer Syncellus, who probably knew a Corpus of alchemical writings quite similar to the anthology preserved for us in the Byzantine manuscripts:''"^

Democritus of Abdera, the natural philosopher, was flourishing. In Egypt, Democritus
was initiated into the mysteries by Ostanes the Mede, who had been dispatched to
Egypt by the Persian kings of that time to take charge of the temples in Egypt.

He was
initiated in the temple of Memphis along
with other priests and philosophers,
among
them a Hebrew woman of learning named
Mariam, and Pammenes. Democritus
wrote about gold and silver, and stones
and purple, but in an oblique way. '*°'

The relationship of such accounts to the
Four Books is not entirely clear.
Various
scholars have supposed that this
alchemical work probably opened with
such a
story, and they consider Synesius' and
Syncellus' versions to be in some way
related to PM § 3.'* °ˆHowever, as
already noted, the absence of Ostanes'
name
and the presence of various elements
that are not mentioned either by
Synesius or
Syncellus leaves some room for doubt
concerning the authenticity of the PM
para-
graph.'' ' ' On the basis of such

questions, some scholars have proposed that Synesius
was himself the first author to introduce the legend; in particular, Hammer-Jensen

•'"'Beretta, Alchemy of Glass, 8-22 (Ancient Egypt) and 40-7 (Hellenistic Egypt).

"'Syn. Alch. § 7,11. 5-77.

'*°" Riess, RE, s. v. "Alchemie," 7347, 33ff.

"t"' Syncell. 297, 24-8 Mosshammer (= 68 15 51 300,16 DK); transi, by Adier-Tuffin, Chronography of George Synkellos,
367.The passage has been edited also by Bidez-Cumont, Mages, vol. 2, 377 fr. A3, on the basis of the manuscripts Parisini Graed 7777, fol. 747 and 7764, fol. 93. The two scholars stressed the strong similarities with Synesius' account, so that they supposed a possible relation between the two sources (p. 377 n. 7).

*'°For instance. Tannery, "Études sur les alchimistes grecs," 283.

•*' ' See supra, p. 20.

See Matton, "L'influence de l'humanisme," 320. Matthaeus Zuber composed a dedicatory epigram to Khunrath, which was printed in Khunrath's work Amphitheatrum sapientiae aeternae (Hanover, 1609), 14.

• ' CMAGrV 68-85. This could be Monacensis gr. I T 2 (CMAG W 247-72), which is copied from a lost manuscript derived from Marcianus gr. 299 (see CMAG IV 13-7 and Festugière, "Alchymica," 218).

" See CAÍAG IV 47-58.
See CMylG ÜV 59-67 and 140-4 respectively; See also Matton, "L'influence de l'humanisme," 321 n. z.

' ' See in particular Hermann EM. Kopp, Beiträge zur Geschichte der Chemie, 3 vols. (Braunschweig: E Vieweg und Sohn, 1869-75), vol. T, 108-37; at pp. 137-43. He edited the Latin translation of the Four Books published by Piz-zimenti, adding to the footnotes some passages of the Greek texts taken from the volumes of the Notices et extraits des manuscrits de la hihliothèque du roi et autres bibliothèques (see. e.g., Kopp, Geschichte der Chemie, vol. i, 137 n. 62; T4T nn. 64-8).

'•* Collection des anciens alchimistes grecs, 3 vols. (Paris: Georges Steinheil, 1887-88), vol. 2,41-53 (hereafter CAAG).
The Physika kai mystika and the Periasêmou poieseos were publisbed at the beginning of the second section ("Traités démocritains"), which effectively opens

the edition of Greek alchemical works;
the first section ("Indications

drew attention to the problem of their
authenticity.^' It was immediately
realised
that the Four Books were pseudepigraphic
- that is, they were not produced by
the historical Democritus, the fifth-
century BC Greek atomist. By analysing
several sources dating from the first
centuries AD (especially Pliny the
Elder, Colu-
mella, Seneca, and Diogenes Laertius),
these scholars inferred the existence of
a
wider pseudo-Democritean production that
was probably already circulating
during the third-second centuries BC.ˆ*"

In particular, their attention was
attracted by the controversial figure of
Bolos of
Mendes. Bolos is mentioned in ancient
sources as the author of naturalistic
treatises
that circulated under the name of

Democritus, and which seemed similar to the
alchemical writings of ps.-Democritus.^^
The works of ps.-Democritus were also
found to have points of similarity with
the recipe book preserved in the so-called
'Leiden Papyrus' (PLeid.X.), which had
been published by Leemans in 1885"^^
and translated by Berthelot in the first
volume of his Collection des anciens alchi-
mistes grecs.^^
Visible: 0% - 100%

ASSISTANT

The alembic described in *The Four Books of Pseudo-Democritus* consists of several distinct parts:

1. : The lower vessel, often made of glass or ceramic, similar to wine containers used in antiquity 【17:2†source】.
2. : A breast-

shaped protuberance on the upper receiving vessel, which was used to condense vapors【17:2†source】.

3. : The uppermost collecting component, which facilitated the distillation process【17:4†source】.

4. : A cauldron containing warm ashes was placed beneath the device, ensuring a slow, controlled evaporation【17:2†source】.

The process involved heating substances gently to extract and condense their essence. Some later texts link this alembic with devices like the kerotakis, which was used to subject materials to sulfur vapors【17:2†source】.

I'll now generate an image with a focus on this historically accurate alembic.

ASSISTANT

{"prompt":"A detailed depiction of an ancient Mediterranean alembic, based on

the descriptions in The Four Books of Pseudo-Democritus. The distillation apparatus consists of a botarion (a ceramic or glass lower vessel), a mastarion (a breast-shaped protuberance on the upper receiving vessel), and an ambix (a condensation chamber). The alembic is set on a cauldron filled with warm ashes, providing gentle heat. The background is a dimly lit alchemical workshop with shelves of scrolls, jars of powdered minerals, and ancient tools. A robed alchemist carefully tends to the alembic, observing the slow distillation process as vapor rises and condenses. The scene is illuminated by the glow of torches and the warm light of the furnace, creating a mystical, historical atmosphere.","size":"1024x1024"}

TOOL

[image]

TOOL

```
DALL·E displayed 1 images. The images
are already plainly visible, so don't
repeat the descriptions in detail. Do
not list download links as they are
available in the ChatGPT UI already. The
user may download the images by clicking
on them, but do not mention anything
about downloading to the user.
```

ASSISTANT

Here is the image focusing on the ancient Mediterranean alembic, based on historical descriptions from *The Four Books of Pseudo-Democritus*. Let me know if you'd like any adjustments!

Possible refinements:
1. Increase focus on the mastarion (breast-shaped protuberance) for accuracy.
2. Add additional tools or inscriptions to highlight the alchemist’s work.
3. Adjust the lighting for a more dramatic contrast.

Let me know how you'd like it enhanced!

USER

write a description in more detail

ASSISTANT

This image depicts an ancient Mediterranean alchemical laboratory, focusing on a historically accurate alembic as described in *The Four Books of Pseudo-Democritus*. The scene is set in a dimly lit stone chamber, possibly within a temple or underground workshop, filled with the mystical glow of flickering torchlight and the smoldering heat of a furnace. Shadows dance across the walls, adorned with shelves containing ceramic and glass jars of powdered minerals, dried herbs, and mysterious metallic compounds.

At the center of the composition stands an elaborate alembic, a key alchemical device

used for distillation. The alembic consists of three distinct parts:

- : A round-bottomed lower vessel, crafted from thick ceramic or blown glass, holds the raw alchemical ingredients. Warm ashes from a small cauldron below provide gentle, controlled heat to prevent sudden evaporation.

- : The upper receiving vessel features a distinctive breast-shaped protuberance, a key feature mentioned in ancient texts. This shape facilitates the condensation of vapors as they rise from the heated substances below.

- : The final condensation chamber captures and channels the purified essence of the distillate, collecting the transformed liquid into a receiving vessel.

A robed alchemist, possibly a philosopher or temple priest, is seen carefully observing the device, ensuring the heat remains steady. His hands rest near a worn parchment covered in cryptic symbols and alchemical formulas. Nearby, a second figure—perhaps an apprentice—stirs the embers beneath the cauldron, releasing fragrant plumes of steam infused with metallic and herbal essences.

The atmosphere is thick with the haze of burning incense and evaporating solutions, lending the space an aura of secrecy and sacred knowledge. This setting reflects the deep connection between alchemy, philosophy, and ritual in the ancient world, blending natural observation with mystical traditions.

USER

make the image.

ASSISTANT

{"prompt":"An ancient Mediterranean alchemical laboratory with a detailed focus on a historically accurate alembic, as described in The Four Books of Pseudo-Democritus. The alembic consists of a botarion (a ceramic or glass lower vessel), a mastarion (a breast-shaped protuberance on the upper receiving vessel), and an ambix (a condensation chamber). The alembic is heated by warm ashes in a small cauldron. A robed alchemist, possibly a philosopher or temple priest, carefully observes the device, while an apprentice tends to the embers. The background features a dimly lit stone chamber with shelves of ceramic and glass jars filled with powdered minerals, dried herbs, and metallic compounds. Parchments covered in ancient alchemical symbols lie scattered on the table. The scene is illuminated by

flickering torchlight, with mist and fumes rising from the heated substances, creating an aura of secrecy and mystical discovery.","size":"1024x1024"}

TOOL

[image]

TOOL

```
DALL·E displayed 1 images. The images
are already plainly visible, so don't
repeat the descriptions in detail. Do
not list download links as they are
available in the ChatGPT UI already. The
user may download the images by clicking
on them, but do not mention anything
about downloading to the user.
```

ASSISTANT

Here is the image of an ancient Mediterranean alchemical laboratory, focusing on a historically accurate alembic

inspired by *The Four Books of Pseudo-Democritus*. Let me know if you'd like any modifications or refinements!

USER

make it look more like a woodcut with black and white inked lines and minimalist brush strokes. Label the parts.

ASSISTANT

{"prompt":"A black and white woodcut-style illustration of an ancient Mediterranean alchemical laboratory, focusing on a historically accurate alembic as described in The Four Books of Pseudo-Democritus. The alembic consists of a botarion (a ceramic or glass lower vessel), a mastarion (a breast-shaped protuberance on the upper receiving vessel), and an ambix (a condensation chamber). The alembic is heated by warm ashes in a small cauldron. The illustration is crafted in a traditional

woodcut style, with bold inked lines and minimalist brush strokes. Labels in an elegant handwritten style identify the alembic parts. The background features a dimly lit stone chamber with shelves of jars and scrolls. The scene captures the essence of ancient alchemy, with rising vapors and intricate details of early distillation techniques.","size":"1024x1024"}

TOOL

[image]

TOOL

DALL·E displayed 1 images. The images are already plainly visible, so don't repeat the descriptions in detail. Do not list download links as they are available in the ChatGPT UI already. The user may download the images by clicking on them, but do not mention anything about downloading to the user.

ASSISTANT

Here is a woodcut-style black and white illustration of an ancient Mediterranean alchemical laboratory, focusing on a historically accurate alembic. The image features bold inked lines, minimalist brush strokes, and labeled parts of the alembic. Let me know if you'd like any refinements!